Hang Li

Postdoctoral Research Fellow | Information Retrieval and Artificial Intelligence

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Academic Profile

Postdoctoral Research Fellow at The University of Queensland with a PhD in Computing Science and expertise in information retrieval, machine learning, large language models and retrieval-augmented generation. Contributes to postgraduate teaching through guest lectures, tutorials, practical support, assessment design and marking, and teaching-team leadership. Principal supervisor of four ongoing master's thesis students and associate supervisor of one ongoing PhD student, with another incoming. Brings experience developing research systems and working in software engineering to teaching that connects AI concepts with implementation, evaluation and professional practice.

Teaching Experience

Guest Lecturer 2025–2026

INFS7410 Information Retrieval

- Delivered 2025 guest lectures on pseudo-relevance feedback, PageRank, dense retrieval and LLM-based feedback methods.
- Connects retrieval theory with current research on neural search, query representations and relevance feedback.
- Guest lecturer for the Semester 2, 2026 postgraduate course, whose curriculum includes neural and LLM retrieval, RAG and agentic search, alongside practical search-system implementation and evaluation.

Teaching Assistant 2025

CSSE7610 Concurrency

- Led weekly tutorials and office hours, supporting students with concurrent programming principles, correctness and Java implementations.
- Contributed to the design and marking of assignments, projects and examinations.

Teaching Assistant 2020–2024

INFS7410 Information Retrieval

- Delivered tutorials and consultations; contributed to assignment, project and examination design and marking.
- Delivered a guest lecture on BERT-based reranking, linking neural retrieval methods to practical search-system development.

Head Teaching Assistant / Teaching Assistant 2020–2022

INFS7202 Web Information Systems

- As Head Teaching Assistant in 2021, organised weekly TA meetings and teaching-team schedules.
- Worked with the course coordinator on syllabus and content design; contributed to assessment design and marking.
- Delivered tutorials and office hours supporting students in web information systems development.

Education

PhD in Computing Science April 2020 – February 2026

The University of Queensland, Brisbane, Australia

- Thesis: Advanced Query Representation and Feedback Methods for Neural Information Retrieval.
- Supervisors: Prof. Guido Zuccon (principal) and A/Prof. Bevan Koopman.
- Research supported by the AgAsk project (UOQ2003-009RTX).

BSc in Computer Science September 2012 – July 2016

University of Minnesota, Twin Cities, Minneapolis, United States

- Global Excellence Scholarship, 2012–2016.

Research Supervision

Master's thesis supervision Principal supervisor of four master's thesis students, all ongoing. Guides research methodology, experimental design and software development across information retrieval, machine learning and large language models.

PhD supervision Associate supervisor of one ongoing PhD student, with another incoming; 40% supervision contribution per student. Provides day-to-day research supervision.

Research mentoring Mentors undergraduate and honours research students, supporting research projects and development of research and technical skills.

Academic and Research Experience

Postdoctoral Research Fellow October 2025 – Present

Information Engineering Lab (IELab), The University of Queensland, Brisbane, Australia

- Develops retrieval-augmented generation systems for clinical question answering, spanning data curation, model fine-tuning, retrieval pipelines, cloud deployment and research prototypes.
- Investigates LLM-based relevance assessment, including overrating by LLM judges and the reliability and cost of automated evaluation.
- Researches agentic search and LLM-based retrieval and reranking; supervises postgraduate research and contributes to collaborative publications.
- Contributes to research funding applications and project reports.

Research Assistant / PhD Researcher December 2020 – February 2026

IELab, The University of Queensland, Brisbane, Australia

- Developed and evaluated pseudo-relevance feedback methods for neural retrieval, examining query representations, feedback quality and combinations of sparse and dense retrievers.
- Contributed to AgAsk, a conversational search system for agricultural questions, and Agvaluate, a test collection for agricultural passage and document retrieval.
- Published in SIGIR, WSDM, ECIR and ACM Transactions on Information Systems; released research code and datasets to support reproducibility.
- Mentored student research projects, presented at international conferences and contributed to project reports and research funding proposals.

Summer Research Scholar November 2019 – March 2020

IELab, The University of Queensland, Brisbane, Australia

- Developed search and recommendation prototypes for systematic review automation, contributing to research on end-to-end query formulation published at SIGIR 2020.

Industry Experience

Software engineering experience in telehealth, e-commerce, financial technology and enterprise IT asset management, covering backend services, RESTful APIs, databases and cloud deployment.

Full Stack Developer July 2018 – July 2019
Henan She Chuan Technology Co., Ltd., Zhengzhou, China

Backend Developer June 2017 – March 2018
Shenzhen Dianmao Technology Group Co., Ltd., Shenzhen, China

Full Stack Developer August 2016 – June 2017
First Capital Fund Management Co., Ltd., Shenzhen, China

Software Developer February 2016 – July 2016
The Velocity Tech Solutions Inc., Minnesota, United States

Research Grant Applications

2026 MRFF Early to Mid-Career Researcher Grant — Co-design and preliminary testing of an AI-enabled tool for personalised dementia risk reduction in Parkinson's disease (PRISM-PD). Role: Chief Investigator. Funding requested: AUD 522,384.30. Status: Submitted; pending review.

Academic Service

Conference organisation Proceedings Chair, CIKM 2021.

Program committees SIGIR, CIKM, ECIR, WSDM and The Web Conference.

Peer review ACM Transactions on Information Systems; ACL Rolling Review.

Awards and Scholarships

2022 SIGIR Student Travel Grant.

2021 Huawei DIGIX Global AI Challenge — team global champion; US$30,000 prize.

2021 ADCS Best Student Paper Award.

2020 SIGIR Student Travel Grant.

2020 UQ Research Training Program Stipend — four-year award.

2019 UQ Dean's Commendation for Academic Excellence.

2012 Global Excellence Scholarship — four-year award.

Teaching Technical and Research Skills

Teaching and supervision Guest lecturing; postgraduate tutorials and consultations; assessment design and marking; teaching-team organisation; research methodology and experimental-design mentoring.

Research methods Retrieval evaluation; reproducible experimentation; query representation and feedback; LLM-based relevance assessment; retrieval-augmented generation.

Programming and machine learning Python, Java, C++, PyTorch, Hugging Face Transformers, PyTerrier, Pyserini, Pandas, NumPy.

Systems and tools Docker, OpenShift, AWS, Linux, Git, MySQL, MongoDB, Redis, LaTeX.

Publications

Journal Articles

J1 AgAsk: An Agent to Help Answer Farmer's Questions from Scientific Documents
Bevan Koopman, Ahmed Mourad, **Hang Li**, Anton van der Vegt, Shengyao Zhuang, Simon Gibson, Yash Dang, David Lawrence, Guido Zuccon
International Journal on Digital Libraries 25(4), pp. 569–584, 2024.

J2 Pseudo Relevance Feedback with Deep Language Models and Dense Retrievers: Successes and Pitfalls
Hang Li, Ahmed Mourad, Shengyao Zhuang, Bevan Koopman, Guido Zuccon
ACM Transactions on Information Systems 41(3), pp. 1–40, 2023.

J3 Design and Research of Intelligent Question-Answering (Q&A) System Based on High School Course Knowledge Graph
Zhijun Yang, Yang Wang, Jianhou Gan, **Hang Li**, Ning Lei
Mobile Networks and Applications, pp. 1–7, 2021.

Conference Papers

C1 Evalugator—Rapid, Agile Development and Evaluation of Retrieval Augmented Generation Systems Without Labels
Bevan Koopman, **Hang Li**, Shuai Wang, Guido Zuccon
ECIR 2026, pp. 63–67.

C2 When LLM Judges Inflate Scores: Exploring Overrating in Relevance Assessment
Chuting Yu, **Hang Li**, Guido Zuccon, Joel Mackenzie, Teerapong Leelanupab
ACM SIGIR 2026, pp. 4322–4327.

C3 AgAsk: A Conversational Search Agent for Answering Agricultural Questions
Hang Li, Bevan Koopman, Ahmed Mourad, Guido Zuccon
ACM WSDM 2023, pp. 1140–1143.

C4 Mesh Suggester: A Library and System for MeSH Term Suggestion for Systematic Review Boolean Query Construction
Shuai Wang, **Hang Li**, Guido Zuccon
ACM WSDM 2023, pp. 1176–1179.

C5 How Does Feedback Signal Quality Impact Effectiveness of Pseudo Relevance Feedback for Passage Retrieval
Hang Li, Ahmed Mourad, Bevan Koopman, Guido Zuccon
ACM SIGIR 2022, pp. 2154–2158.

C6 To Interpolate or Not to Interpolate: PRF, Dense and Sparse Retrievers
Hang Li, Shuai Wang, Shengyao Zhuang, Ahmed Mourad, Xueguang Ma, Jimmy Lin, Guido Zuccon
ACM SIGIR 2022, pp. 2495–2500.

C7 Pseudo-Relevance Feedback with Dense Retrievers in Pyserini
Hang Li, Shengyao Zhuang, Xueguang Ma, Jimmy Lin, Guido Zuccon
ADCS 2022, pp. 1–6.

C8 Improving Query Representations for Dense Retrieval with Pseudo Relevance Feedback: A Reproducibility Study
Hang Li, Shengyao Zhuang, Ahmed Mourad, Xueguang Ma, Jimmy Lin, Guido Zuccon
ECIR 2022, pp. 599–612.

C9 Implicit Feedback for Dense Passage Retrieval: A Counterfactual Approach
Shengyao Zhuang, **Hang Li**, Guido Zuccon
ACM SIGIR 2022, pp. 18–28.

C10 MeSH Term Suggestion for Systematic Review Literature Search
Shuai Wang, **Hang Li**, Harrisen Scells, Daniel Locke, Guido Zuccon
ADCS 2021, pp. 1–8.

Publications Continued

Conference Papers Continued

C11 Deep Query Likelihood Model for Information Retrieval
Shengyao Zhuang, **Hang Li**, Guido Zuccon
ECIR 2021, pp. 463–470.

C12 Systematic Review Automation Tools for End-to-End Query Formulation
Hang Li, Harrisen Scells, Guido Zuccon
ACM SIGIR 2020, pp. 2141–2144.

TREC Track Papers

T1 IELAB at TREC Deep Learning Track 2021
Shengyao Zhuang, **Hang Li**, Shuai Wang, Guido Zuccon
Proceedings of the 2021 TREC Deep Learning Track, 2021.

T2 IELAB for TREC Conversational Assistance Track (CAsT) 2020
Sebastian Cross, **Hang Li**, Arvin Zhuang, Ahmed Mourad, Bevan Koopman, Guido Zuccon
Proceedings of the 29th Text REtrieval Conference (TREC 2020), 2020.

Datasets

D1 Agvaluate: A New Test Collection for Both Passage and Document Retrieval in the Agriculture Domain
Hang Li, Ahmed Mourad, Bevan Koopman, Guido Zuccon
Research dataset, The University of Queensland, 2022.

Preprints and Submitted Manuscripts

P1 Effective Dense Retrieval using Only In-Context Examples
Nour Jedidi, Abdul Basit Ali, **Hang Li**, Jimmy Lin
Submitted to NAACL 2027, 2026.

P2 AOBA: Adaptive Overlapping Block Aggregation for Efficient LLM Reranking
Hang Li, Bevan Koopman, Guido Zuccon
Submitted to ECIR (submission in 2026).

P3 Whole-Pool Setwise Reranking with Long-Context Language Models
Hang Li, Chuting Yu, Teerapong Leelanupab, Bevan Koopman, Guido Zuccon
Submitted to ECIR (submission in 2026).

P4 Bridging the Gap Between Small and Large Dense Retrievers via Offline Generative Feedback
Hang Li, Chuting Yu, Xiao Wang, Bevan Koopman, Guido Zuccon
Submitted to ACM Transactions on Information Systems, 2026.

P5 A Survey of Late-Interaction Neural Retrieval: Paradigms, Systems, and Research Frontiers
Xiao Wang, Chuting Yu, Minghan Li, Binci Yang, **Hang Li**, Ben He
Submitted to Information Sciences, 2026.

P6 Pseudo-Relevance Feedback Can Improve Zero-Shot LLM-Based Dense Retrieval
Hang Li, Xiao Wang, Bevan Koopman, Guido Zuccon
Preprint, 2025. arXiv: 2503.14887v1 [cs.IR].

P7 LLM-VPRF: Large Language Model Based Vector Pseudo Relevance Feedback
Hang Li, Shengyao Zhuang, Bevan Koopman, Guido Zuccon
Preprint, 2025. arXiv: 2504.01448 [cs.IR].

P8 TPRF: A Transformer-based Pseudo-Relevance Feedback Model for Efficient and Effective Retrieval
Hang Li, Chuting Yu, Ahmed Mourad, Bevan Koopman, Guido Zuccon
Preprint, 2024. arXiv: 2401.13509 [cs.IR].